Water Filtration Unit



Category: **Hospitality & Housekeeping** Type: **Full SOP** Priority: **Medium** 22 labs

Covers these lab items: Filtering machines; Filtration System; Mini water filtration unit

What it is used for

A water filtration unit removes selected suspended particles, impurities, or other contaminants from water using suitable filter media and treatment stages. Correct operation and timely filter replacement help maintain consistent water quality, but filtration alone does not guarantee drinking-water safety.

Main parts

- Raw-water inlet and isolation valve
- Filter housing and filter cartridges or media
- Pre-filter and sediment filter where fitted
- Activated carbon, membrane, or other treatment stage as designed
- Treated-water outlet and storage tank if fitted
- Pressure gauge, flow indicator, and drain connection where fitted

Before you start

1. Read the manufacturer's instructions and confirm the unit's intended water source and treatment capability.
2. Inspect the housing, pipes, valves, seals, electrical connections, and fittings for damage or leakage.
3. Verify that filter cartridges and media are approved for the model and are within their service life.
4. Confirm the raw-water supply pressure and quality are within the unit's operating limits.
5. Flush new filters according to the manufacturer's instructions before collecting treated water.
6. Check that the treated-water tank and outlet are clean and that required water-quality testing is current.

How to use

1. Open the raw-water inlet valve according to the operating instructions.
2. Start the pump or powered treatment stage if the unit requires it.
3. Check the pressure and flow indicators and inspect the system for leaks.
4. Allow the specified startup flush to pass through the filters and send it to the approved drain if required.
5. Collect treated water only after the flushing procedure is complete.
6. Monitor flow rate, pressure drop, unusual odour, colour, or other signs of reduced performance.
7. Record operating readings and water-quality checks according to the facility procedure.

8. Close the inlet valve and isolate power before replacing filters or opening the housing.
9. Replace exhausted cartridges or media using the approved procedure and reseal the housing correctly.
10. Flush the unit after filter replacement and verify that it operates normally before returning it to service.

⚠ Safety

- Depressurize the filter housing before loosening its cap or removing cartridges.
- Use only filter media and replacement parts approved for the unit and water application.
- Disconnect electrical power before servicing pumps, controls, or electrical components.
- Do not drink treated water unless its quality meets the applicable drinking-water requirements.
- Handle chemical cleaning or disinfection agents according to their safety instructions.
- Repair leaks promptly and keep electrical equipment away from standing water.

Care & storage

- Replace cartridges and filter media according to the manufacturer's schedule or verified performance limits.
- Clean and sanitize the housing and storage tank at the specified intervals.
- Inspect valves, seals, pipes, pressure indicators, and connections regularly.
- Maintain records of filter changes, flushing, servicing, and water-quality tests.

Common mistakes to avoid

- Delaying filter replacement can reduce flow and allow treatment performance to deteriorate.
- Failing to flush new filters can leave manufacturing residues or loose media in the water.
- Using incompatible cartridges can damage the unit or reduce contaminant removal.
- Assuming clear-looking water is automatically safe can expose users to invisible contaminants.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.